

Real-Guided Skeleton Augmentation for Actuator-Constrained Rock-Paper-Scissors Robot Counterattack

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Abstract—This report presents a skeleton-first real-to-sim-to-real pipeline for an embodied Rock-Paper-Scissors counterattack task. The project begins with a small set of real hand-motion videos, extracts MediaPipe 21-landmark skeletons, canonicalizes the trajectories, and expands them into real-guided synthetic skeleton datasets aligned to SCHUNK/Isaac-style robot-hand pose progress. From only 20 reviewed real clips and 720 processed frames, the pipeline builds a 10,000-sample sharded skeleton dataset and a render-ready 32-frame alignment layer. The final live validation applies the frozen v4 temporal predictor back to a prompt-window camera capture and produces an actuator-feasible robot counterattack with 0.4667 s remaining. The robot hand is therefore the final embodied response target, while the main research contribution is the real-guided skeleton augmentation and transfer path.

Index Terms—real-to-sim, sim-to-real, skeleton sequence modeling, MediaPipe, early action prediction, actuator feasibility, robot hand

I. INTRODUCTION

Embodied interaction requires more than recognizing a final pose after an action is complete. In a Rock-Paper-Scissors counterattack, a robot must infer the human’s intended gesture early enough to choose a winning response and still move its hand before the actuator deadline. This makes the problem a combined data, prediction, and control problem: sparse real hand-motion data must support an early temporal predictor, and the predictor must transfer back to live camera input under timing constraints.

The main difficulty in this project is data scarcity. Recording many high-quality real hand-motion videos is expensive, and final-frame pose labels do not capture when the hand begins to reveal the final gesture. The project therefore uses a skeleton-first real-to-sim-to-real strategy. A small set of real videos is converted into MediaPipe 21-landmark trajectories, those trajectories are canonicalized into hand-local coordinates and temporal features, and real-guided augmentation generates broader skeleton motion coverage. The synthetic skeletons are aligned to SCHUNK/Isaac-style pose progress so they remain connected to the intended robot-hand embodiment.

Fig. 1 summarizes the final pipeline. The pipeline is deliberately not a raw image-to-landmark Isaac perception system. Instead, it follows the project boundary of skeleton ground truth first: camera frames are used to obtain MediaPipe skeletons,

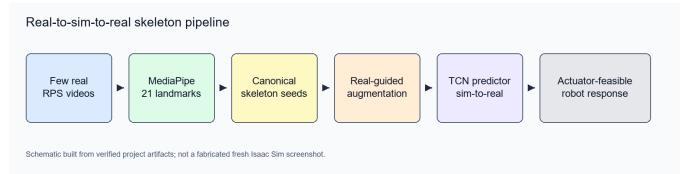


Fig. 1. Real-to-sim-to-real pipeline schematic. The figure is a schematic built from verified project artifacts, not a fresh Isaac Sim screenshot.

and the temporal model consumes canonical skeleton features. The robot hand appears at the end as an actuator-constrained response target.

This report makes four contributions:

- a few-shot real RPS skeleton seed dataset extracted from MediaPipe 21-landmark hand tracks;
- a real-guided skeleton augmentation pipeline with compact and large sharded datasets;
- a render-ready SCHUNK/Isaac alignment layer that improves pose-progress quantization over the earlier 8-frame grid;
- a sim-to-real prompt-window counterattack demo using the frozen v4 predictor and actuator feasibility checks.

II. RELATED WORK

Early action prediction differs from ordinary action recognition because the model must decide before the full motion is observed [1], [2]. This project narrows that setting to prompt-conditioned hand motion, where useful evidence is distributed across the transition from a starting fist to the final RPS pose. The final-frame class is not sufficient: the decision must arrive early enough for a robot hand to respond.

Skeleton-based temporal models are a practical fit for this setting. GRU and TCN-style models can process short sequential feature vectors without requiring a full raw-video perception stack [3]. This is why the project uses MediaPipe to extract 21 landmarks and then trains on canonical skeleton coordinates, velocities, finger-curl values, and motion signals. Selective prediction and calibration ideas also matter, because the system should not issue a counterattack when confidence, margin, or confirmation gates are weak [4], [5].

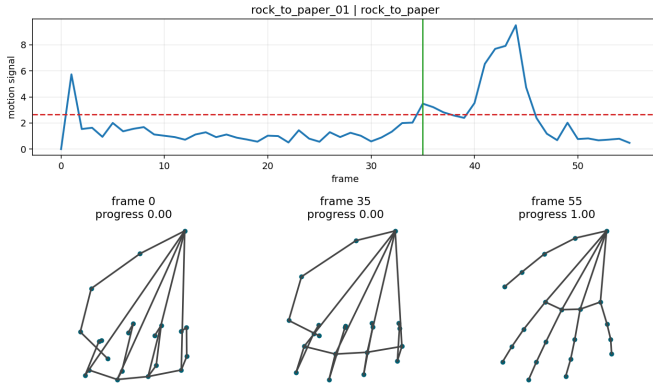


Fig. 2. Canonical MediaPipe skeleton seed example. Real videos are converted into hand-local skeleton trajectories with motion signal, onset, and progress metadata.

Isaac Sim and Isaac Lab provide the target context for articulated robot assets and actuator modeling [6], [7]. In this submission, Isaac/SCHUNK work is used as render and robot-response grounding. The final learning path remains skeleton-first, while the SCHUNK-style hand is used to represent the robot response and to check whether a selected counter gesture is actuator feasible.

III. METHOD

A. Real MediaPipe Skeleton Seeds

The first stage records a small real-video seed set and extracts MediaPipe Hands landmarks. The verified seed review contains 20 real clips, split evenly between `rock_to_paper` and `rock_to_scissors`. MediaPipe returned 21 landmarks for every processed frame, giving 720 processed frames and minimum detection coverage of 1.0. Multiple-hand detections were preserved in metadata, but the highest-confidence hand was used as the primary hand.

B. Canonical Skeleton Features

The extracted landmarks are converted into canonical hand-local coordinates. The wrist is the origin, wrist-to-middle-MCP distance defines the primary scale, and the palm axes define a local coordinate frame. Each frame stores canonical 21-point coordinates, velocities, five finger-curl values, five tip-to-MCP distances, five fingertip speeds, and one motion signal. The resulting feature vector has dimension 142. Fig. 2 shows a representative canonical seed preview with motion signal and progress checkpoints.

C. Real-to-Sim Skeleton Augmentation

The canonical real skeleton seeds drive the synthetic expansion stage. A compact bulk dataset generates 2,000 augmented samples, balanced between `rock_to_paper` and `rock_to_scissors`. A larger sharded dataset generates 10,000 samples with a 7,000/1,500/1,500 train/validation/test split. The augmentation is intentionally real-guided: temporal

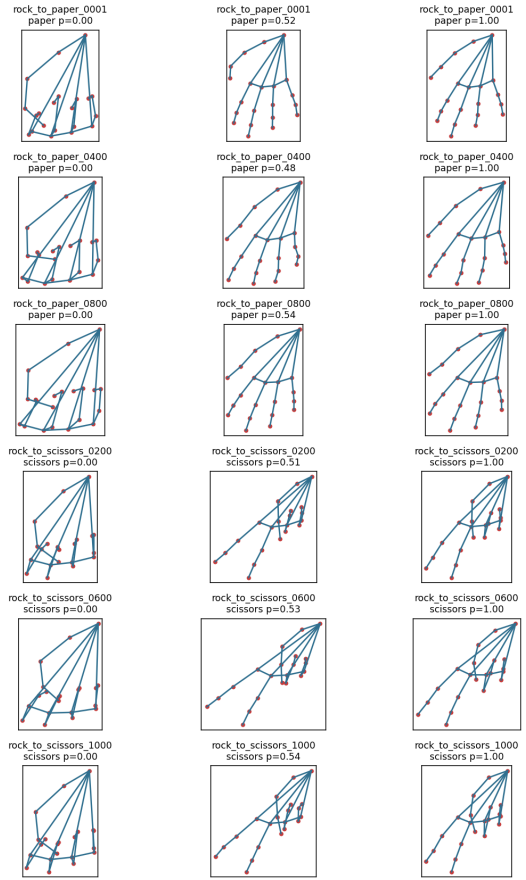


Fig. 3. Real-guided skeleton augmentation evidence. Few real MediaPipe skeleton seeds are expanded into multiple canonical skeleton trajectories for `rock_to_paper` and `rock_to_scissors`.

scaling, temporal warping, finger-specific scaling, small rotations, and landmark noise are applied to canonical skeleton trajectories rather than to arbitrary hand drawings.

D. SCHUNK/Isaac Render-Ready Alignment

The augmented skeleton sequences are aligned to a SCHUNK/Isaac-style robot-hand pose schedule. The older connected-motion evidence used an 8-frame progress grid; the real-guided render-ready alignment prepares a 32-frame schedule with 64 future render-manifest entries, 32 for each transition. This stage does not claim that every sample has a fresh rendered image. Instead, it stores per-frame lookup arrays that make the skeleton dataset ready to consume future SCHUNK render records.

Fig. 4 shows the alignment improvement. The maximum motion-progress quantization error drops from about 0.106 with the old 8-frame grid to about 0.016 with the 32-frame real-guided schedule.

E. Sim-to-Real Counterattack

After training and validation iterations, the final live/demo predictor is frozen to v4. It uses the rock-guard realtime selector config, profile weights $[0.25, 0.75, 0.0]$, confidence

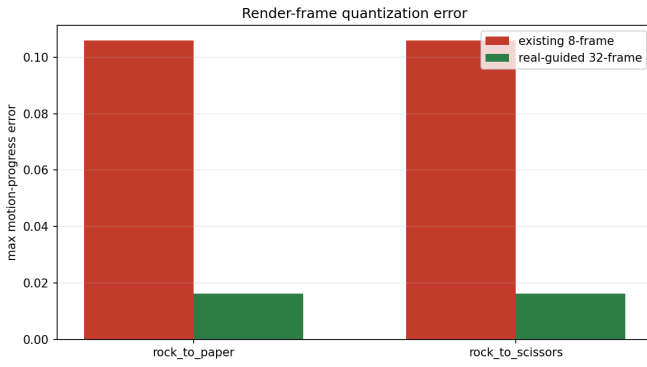


Fig. 4. Render-ready alignment improvement. The 32-frame real-guided schedule reduces the maximum motion-progress quantization error relative to the earlier 8-frame SCHUNK progress grid.

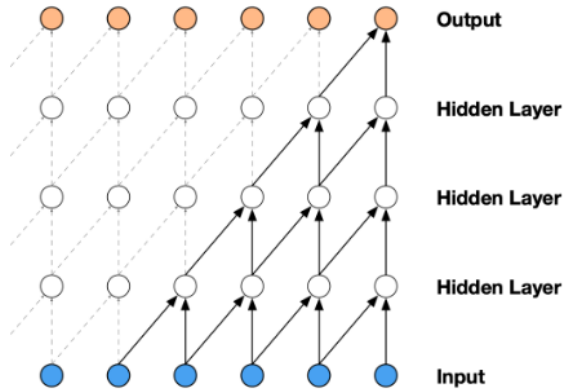


Fig. 5. Temporal predictor schematic. The model consumes a sequence of canonical skeleton features and predicts the response-window gesture before the robot-hand deadline.

threshold 0.70, margin threshold 0.10, confirmation count 2, and response prompt `scissors`. `v7e` is retained as diagnostics only and is not promoted. Fig. 5 shows the temporal convolution style of the sequence predictor as a conceptual model diagram, not as experimental evidence by itself.

The live prompt sequence is:

`rock → paper → scissors`.

The human performs the target gesture during the final `scissors` response window. The counter policy maps human `rock` to robot `paper`, human `paper` to robot `scissors`, and human `scissors` to robot `rock`. The robot starts from `rock`, waits for a confirmed response-window prediction, and then checks response delay, deadline, velocity, and normalized joint limits before accepting the episode as actuator feasible.

IV. EXPERIMENTS

Table I summarizes the real-to-sim dataset path. The key point is that the project does not jump directly from a few real videos to a final demo. It first constructs a verified skeleton representation, then expands it into larger real-guided synthetic datasets.

TABLE I
REAL-GUIDED SKELETON DATASET PIPELINE

Stage	Size	Evidence
MediaPipe seed review	20 clips, 720 frames	min detection coverage 1.0
Canonical seed dataset	20 sequences	max length 56, feature dim 142
Compact augmentation	2,000 samples	1,000 per transition
Large sharded dataset	10,000 samples	split 7,000 / 1,500 / 1,500
Render-ready alignment	64 manifest entries	32 frames per transition

TABLE II
SCHUNK/ISAAC RENDER-READY ALIGNMENT

Alignment grid	Max progress error	Role
Existing 8-frame grid	about 0.106	coarse connected-motion reference
Real-guided 32-frame grid	about 0.016	render-ready alignment layer

Table II summarizes the alignment improvement. This table is important because it connects skeleton augmentation to the later robot-hand embodiment. The large skeleton dataset remains usable for skeleton-only training, while the render-ready fields make the same samples compatible with future SCHUNK/Isaac render-conditioned experiments.

V. FINAL LIVE VALIDATION

The final live validation tests whether the skeleton predictor transfers back to a real prompt-window camera capture. The selected final take is `human rock → robot paper`. During the final `PROMPT SCISSORS` response window, the system predicts `rock` and issues `robot paper`. The actuator feasibility checker reports an actuator-feasible win with 0.4667 s remaining. Fig. 6 shows the poster frame.

The robot-hand visuals use archived SCHUNK-style render keyframes as the response pose library, shown in Fig. 7. These images explain the screen-rendered robot response, but they are not the main evidence for the real-guided data expansion.

Table III records the final model and demo policy. The `v4` fallback remains the live/demo predictor. `v7e` reached original20 strict validation at 17/20 and is retained only as diagnostic evidence; it is not promoted and `v7f` retraining is not started.

VI. DISCUSSION

The most important result is not that a screen-rendered robot hand appears in the final video. The important research path is that a small real skeleton seed set can be converted into a larger real-guided skeleton dataset, aligned to the robot-hand embodiment, and then transferred back to live camera input for a timing-constrained response.

The project also exposes a limitation. The final training and live/demo path is skeleton-first. The render-ready sidecars prepare the dataset for image- or render-conditioned Isaac experiments, but the final predictor does not learn from raw Isaac camera images. This is an intentional `v1` boundary rather than a hidden failure. It keeps the final system verifiable and avoids confusing skeleton prediction with a separate vision-based landmark-estimation problem.

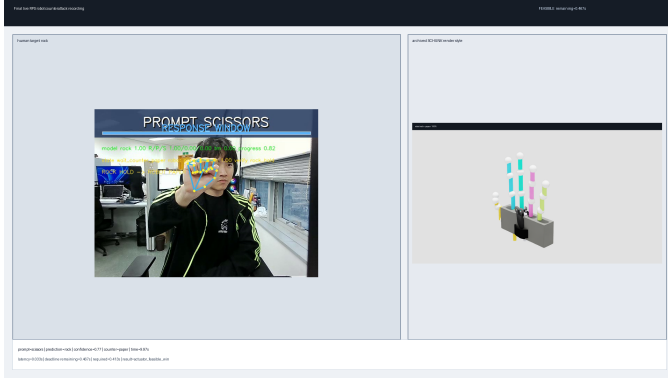


Fig. 6. Final sim-to-real counterattack validation. The frozen skeleton predictor is applied back to live prompt-window input and drives an actuator-feasible robot paper response.



Fig. 7. Archived SCHUNK-style response pose library for rock, paper, and scissors. These verified keyframes support the screen-rendered robot response layer.

The counterattack demo remains valuable as an embodied validation target. It forces the temporal predictor to be useful before the prompt window closes and rejects correct but late decisions through actuator feasibility. The failed diagnostic takes are therefore part of the evidence: a model can fail by making no stable decision, by selecting the wrong class, or by deciding too late for the robot to move.

VII. CONCLUSION

This project implements a real-guided skeleton augmentation pipeline for actuator-constrained RPS counterattack. It starts from 20 real MediaPipe-reviewed clips, builds canonical skeleton seeds, expands them into compact and 10,000-sample real-guided datasets, aligns the data to SCHUNK/Isaac-style pose progress, and validates the frozen v4 predictor back on live prompt-window input. The final robot-hand counterattack is the application-level proof that the prediction and timing policy can drive an actuator-feasible response, while the dataset and alignment pipeline form the central research contribution.

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TABLE III
FINAL PREDICTOR AND LIVE DEMO METRICS

Category	Metric	Value
v4 fallback	profile weights	[0.25, 0.75, 0.0]
v4 fallback	original20 strict	20/20
v4 fallback	heldout15 validation	11/15, validation-only
v7e diagnostics	original20 strict	17/20, not promoted
Final live take	selected take	human rock - robot paper
Final live take	episode result	actuator-feasible win
Final live take	remaining time	0.4667 s

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